

CALCULUS (II)

WEEK (VII)

QUADRATIC SURFACES

→ The most general second degree equation in three variables is

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz = J.$$

→ If the above equation can be factored in the form

$$(A_1x + B_1y + C_1z - D_1)(A_2x + B_2y + C_2z - D_2) = 0$$

degenerate case

→ Then, the graph is a pair of planes

$$A_1x + B_1y + C_1z = D_1 \text{ and } A_2x + B_2y + C_2z = D_2$$

or one plane if the two linear equations represent the same plane.

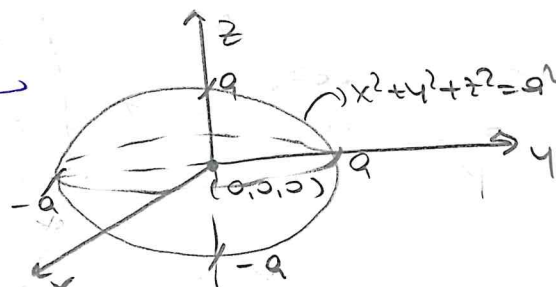
→ If such factorization is not possible then the surface represented by the general second-degree equation is called a quadratic surface.

Spheres The equation $x^2 + y^2 + z^2 = a^2$ represents a sphere of radius a centered at the origin. More generally,

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

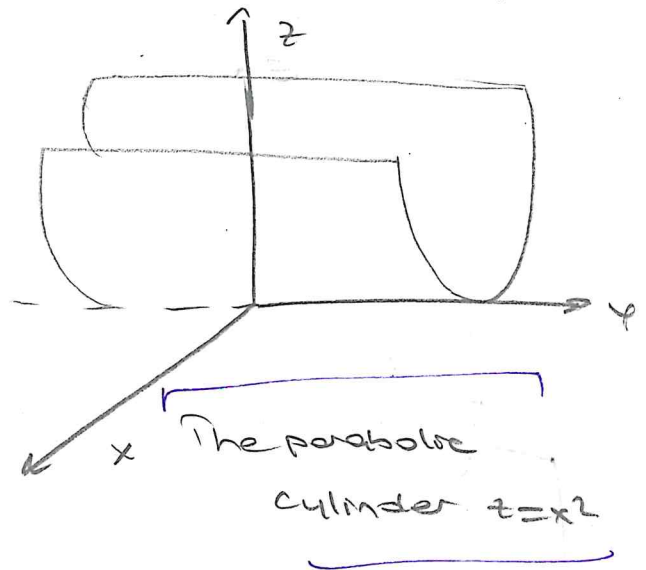
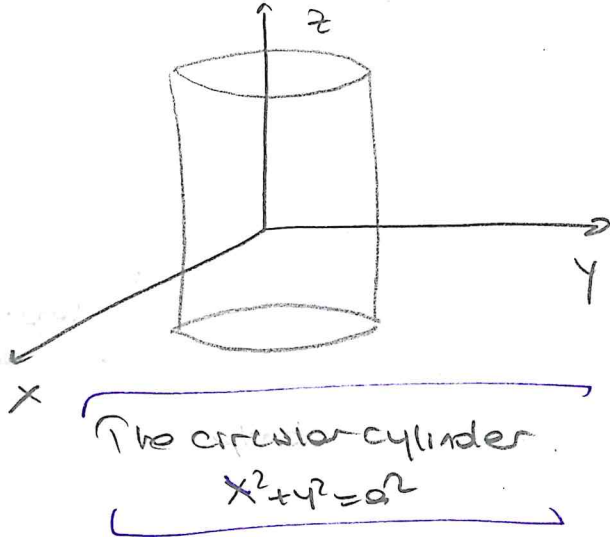
represents a sphere of radius a centered at the point (x_0, y_0, z_0) . If a quadratic equation in x, y and z has equal coefficients for x^2, y^2, z^2

and has no other second-degree terms, then it will represent if any surface at all, a sphere. The center can be found by completing the squares for circles in the plane

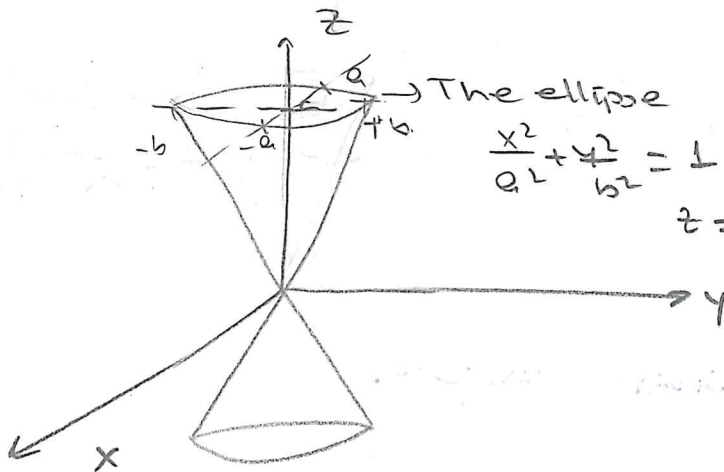


Cylinders

A cylinder is a surface that is generated by moving a straight line along a planar curve while holding the line parallel to a given fixed line. The curve is called a generating curve for the cylinder.



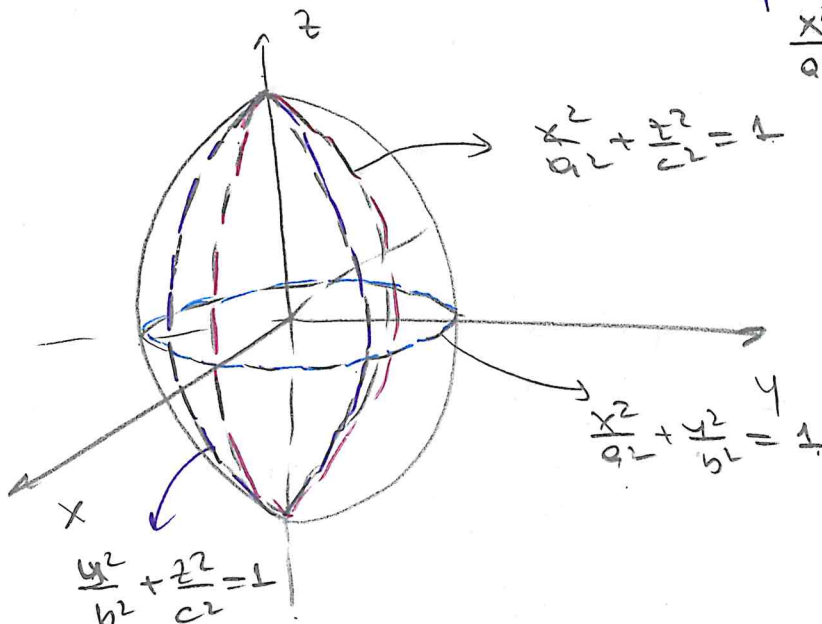
Cones



Elliptic Cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$$

Ellipsoids

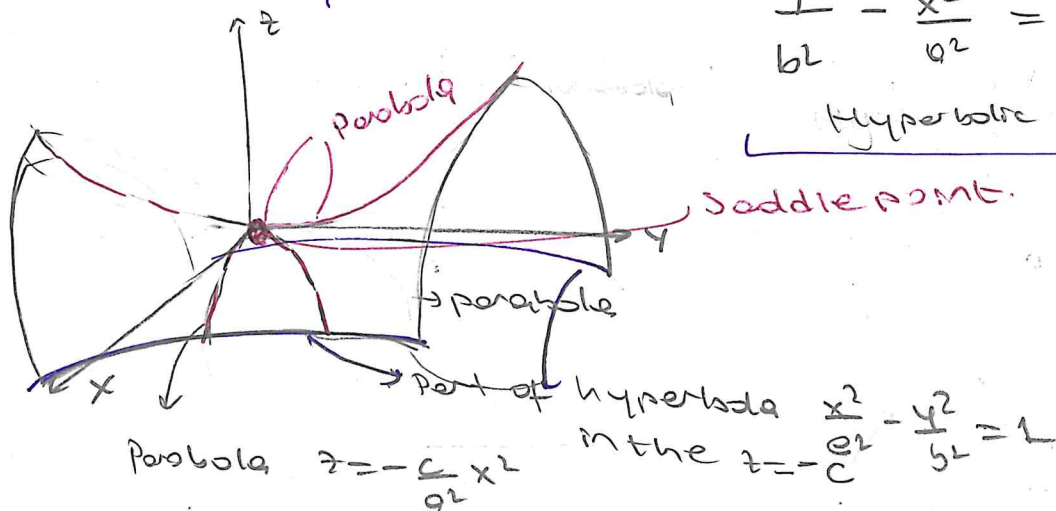


$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Ellipsoid

Paraboloids

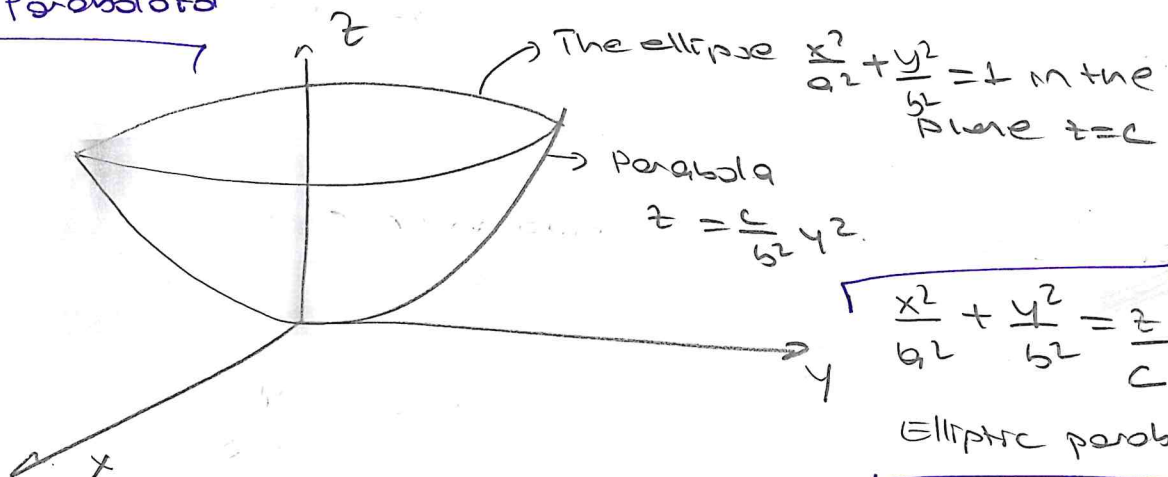
Hyperbolic Paraboloid



$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c} \quad c > 0$$

Hyperbolic paraboloid

Elliptic Paraboloid

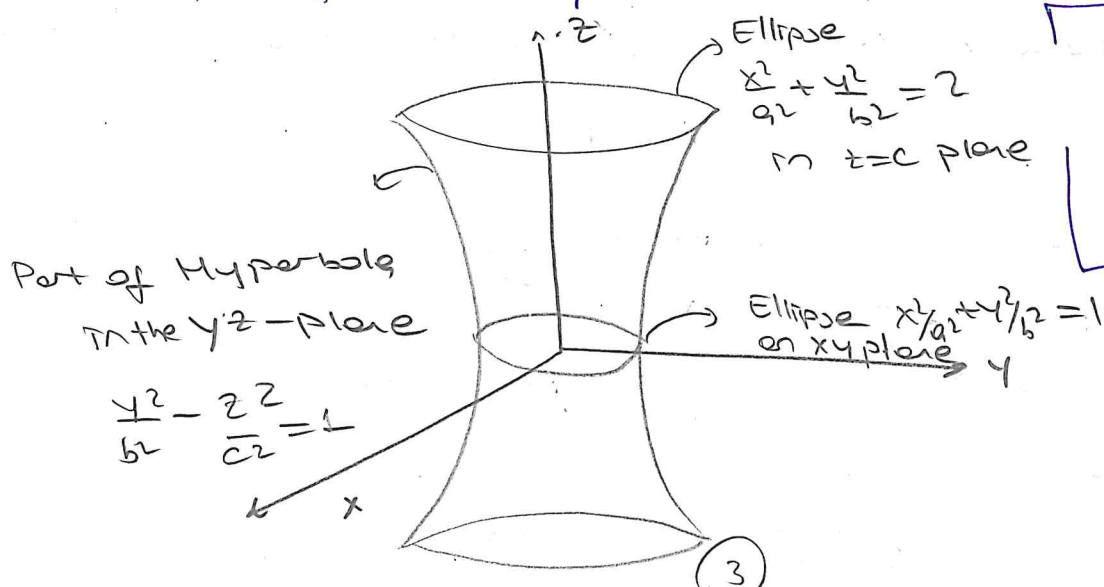


$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$

Elliptic paraboloid

Hyperboloids

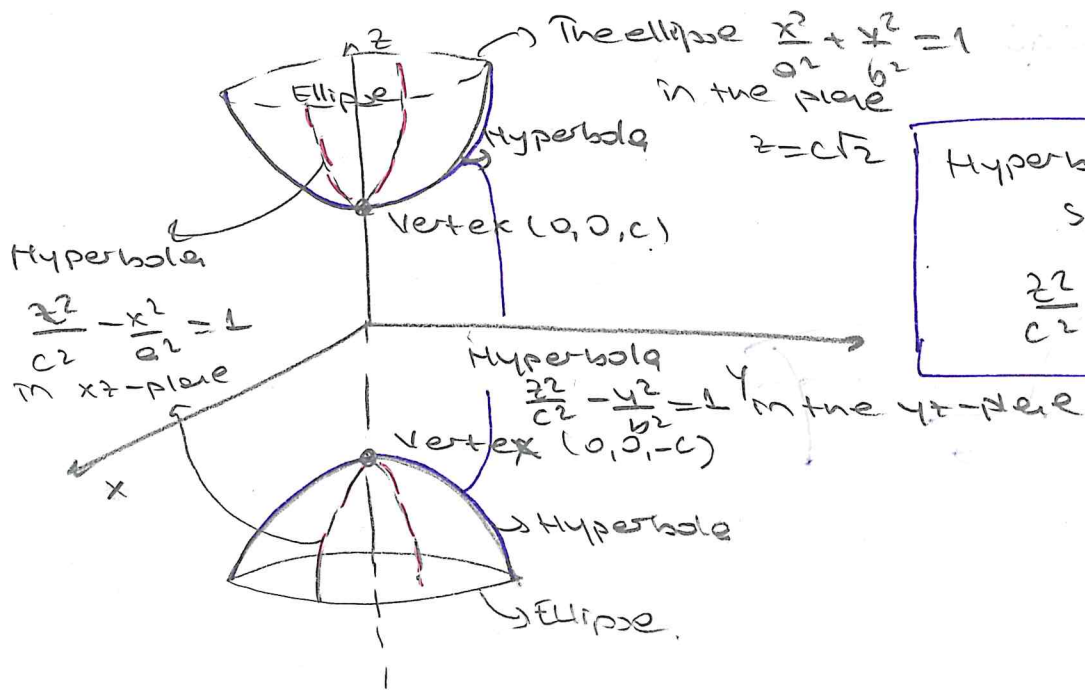
Hyperboloid of one sheet



Hyperboloid of one sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Hyperboloid of two sheets



Hyperboloid of two
Sheets

$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

VECTOR FUNCTIONS OF ONE VARIABLE

Vector Function A function of having vector values.

Our main interest is to study differential and integral calculus of vector functions of a single variable

such functions:

- can be used to represent curves parametrically;
- can be thought of as giving the position, at time t , of a moving point (or a particle)

→ If a particle moves around in 3-space, its motion can be described by giving the three coordinates of its position as functions of time t :

$$x = x(t), \quad y = y(t) \quad \text{and} \quad z = z(t)$$

→ It is more convenient, however, to replace these three equations by a single vector equation,

$\mathbf{r} = \mathbf{r}(t)$ giving the position vector of the moving particle as a function of t .

→ In terms of the standard basis vectors $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$, the position of the particle at time t is

$$\text{position: } \mathbf{r} = \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

$$(\text{in 2-space } \mathbf{r} = \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j})$$

Domain The domain of a vector function is the set of all t 's for which all the component functions are defined.

Definition Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be a vector function

with domain D , and $\mathbf{L}$ is a vector. We say that $\mathbf{r}$

has limit $\mathbf{L}$ as t approaches t_0 and write

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L}$$

If for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$|\mathbf{r}(t) - \mathbf{L}| < \epsilon \quad \text{whenever} \quad 0 < |t - t_0| < \delta$$

→ If $\vec{L} = L_1 \vec{i} + L_2 \vec{j} + L_3 \vec{k}$, then it can be shown that

$\lim_{t \rightarrow t_0} \vec{r}(t) = \vec{L}$ precisely when

$$\lim_{t \rightarrow t_0} f(t) = L_1, \quad \lim_{t \rightarrow t_0} g(t) = L_2 \quad \text{and} \quad \lim_{t \rightarrow t_0} h(t) = L_3.$$

We omit the proof. The equation

$$\lim_{t \rightarrow t_0} \vec{r}(t) = \left(\lim_{t \rightarrow t_0} f(t) \right) \vec{i} + \left(\lim_{t \rightarrow t_0} g(t) \right) \vec{j} + \left(\lim_{t \rightarrow t_0} h(t) \right) \vec{k}$$

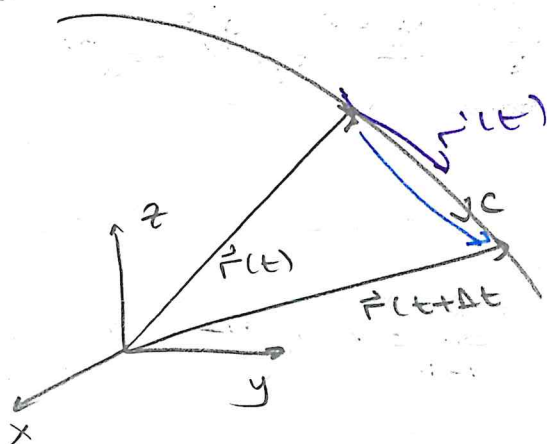
provides a practical way to calculate limits of vector functions.

Ex 7 If $\vec{r}(t) = \cos t \vec{i} + \sin t \vec{j} + t \vec{k}$ then

$$\begin{aligned} \lim_{t \rightarrow \pi/4} \vec{r}(t) &= \left(\lim_{t \rightarrow \pi/4} \cos t \right) \vec{i} + \left(\lim_{t \rightarrow \pi/4} \sin t \right) \vec{j} + \left(\lim_{t \rightarrow \pi/4} t \right) \vec{k} \\ &= \frac{\sqrt{2}}{2} \vec{i} + \frac{\sqrt{2}}{2} \vec{j} + \frac{\pi}{4} \vec{k} \end{aligned}$$

Definition A vector function $\vec{r}(t)$ is continuous at a point $t=t_0$ in its domain if $\lim_{t \rightarrow t_0} \vec{r}(t) = \vec{r}(t_0)$. The function is continuous

if it is continuous at every point in its domain.



$$\text{Average velocity} = \frac{\vec{r}(t+\Delta t) - \vec{r}(t)}{\Delta t}$$

$$\text{velocity } \vec{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t+\Delta t) - \vec{r}(t)}{\Delta t} = \frac{d}{dt} \vec{r}(t)$$

(If the limit exists, $\vec{r}$ is called differentiable at t)

$$\text{Speed} = |\vec{v}(t)|$$

Ex 7 $\lim_{t \rightarrow 1} \vec{r}(t) = ?$ $\vec{r}(t) = t^3 \vec{i} + \frac{\sin(3t-3)}{t-1} \vec{j} + e^{2t} \vec{k}$

function is undefined at $t=1$.

$$\lim_{t \rightarrow 1} t^3 = 1$$

$$\lim_{t \rightarrow 1} \frac{\sin(3t-3)}{(t-1)} = \lim_{t \rightarrow 1} \frac{\cos(3t-3) \cdot 3}{1} = \lim_{t \rightarrow 1} \cos(3t-3) \cdot 3 = 3$$

$$\lim_{t \rightarrow 1} e^{2t} = e^2$$

The rules for addition and scalar multiplication of vectors imply that

$$V(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \left(\frac{x(t+\Delta t) - x(t)}{\Delta t} \vec{i} + \frac{y(t+\Delta t) - y(t)}{\Delta t} \vec{j} + \frac{z(t+\Delta t) - z(t)}{\Delta t} \vec{k} \right)$$

$$\mathbf{r}'(t) = \frac{dx}{dt} \vec{i} + \frac{dy}{dt} \vec{j} + \frac{dz}{dt} \vec{k}$$

Thus the vector function $\mathbf{r}$ is differentiable at t if and only if its three components x, y and z are differentiable at t .

Thm Derivative The vector function $\mathbf{r}(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k}$ has a derivative (is differentiable) at t if f, g, h have derivatives.

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{df}{dt} \vec{i} + \frac{dg}{dt} \vec{j} + \frac{dh}{dt} \vec{k}$$

→ The vector $\mathbf{r}'(t)$ is called the tangent vector to the curve defined by $\mathbf{r}(t)$ at the point P , provided that $\mathbf{r}'(t)$ exists and $\mathbf{r}'(t) \neq \mathbf{0}$.

Unit tangent vector $\vec{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$

Ex 1) Find the derivative of $\vec{r}(t) = (1+t^3)\vec{i} + te^{-t}\vec{j} + \sin 2t\vec{k}$

2) Find the unit tangent vector at point where $t=0$.

a) $\mathbf{r}'(t) = 3t^2\vec{i} + (e^{-t} - te^{-t})\vec{j} + 2\cos(2t)\vec{k}$

b) $\mathbf{r}'(0) = 0\vec{i} + \vec{j} + 2\vec{k} = \langle 0, 1, 2 \rangle$

$$\vec{T}(t) = \frac{\mathbf{r}'(0)}{|\mathbf{r}'(0)|} = \frac{0\vec{i} + \vec{j} + 2\vec{k}}{\sqrt{1+4}} = \left\langle 0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$$

The point $\vec{r}(0) = \vec{i} + 0\vec{j} + 0\vec{k} = \vec{i} = (1, 0, 0)$. $\vec{T}(t)$ is the tangent vector at point $\mathbf{r}(0)$.

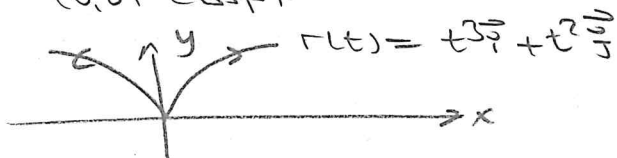
Smooth Curve A smooth curve is any curve for which $\mathbf{r}'(t)$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$ for any t except possibly at the end points.

Ex 7 Consider the plane curve $r(t) = t^3 \vec{i} + t^2 \vec{j}$

$r(t)$ is continuous since t^3, t^2 have continuous derivatives of all orders. However, curve is not smooth at the origin

($x = t^3$ $y = t^2$ $y = x^{2/3}$ (0,0) cusp)

$$v(t) = 3t^2 \vec{i} + 2t \vec{j}$$



$r(t)$ Curve is smooth for all points except $v(t) \neq 0$.

Def Acceleration Acceleration of the particle is the time derivative of the velocity.

$$\text{acceleration } a(t) = \frac{dv}{dt} = \frac{d^2 r}{dt^2}$$

Ex 7 Describe the curve $r(t) = t \vec{i} + t^2 \vec{j} + t^3 \vec{k}$. Find velocity and acceleration for this curve at $(1,1,1)$

$$v(t) = \frac{dr}{dt} = \vec{i} + 2t \vec{j} + 3t^2 \vec{k}$$

$$v(t) \Big|_{(1,1,1)} = \vec{i} + 2\vec{j} + 3\vec{k}$$

$$a(t) = \frac{dv}{dt} = \frac{d^2 r}{dt^2} = 2\vec{j} + 6t \vec{k}$$

$$a(t) \Big|_{(1,1,1)} = 2\vec{j} + 6\vec{k}$$

THM DIFFERENTIATION RULES FOR VECTOR FUNCTIONS

Let $u(t)$ and $v(t)$ be differentiable vector-valued functions, and let $\lambda(t)$ be a differentiable scalar-valued function. Then,

(a) $\frac{d}{dt} (u(t) + v(t)) = u'(t) + v'(t)$ (b) $\frac{d}{dt} (\lambda(t) u(t)) = \lambda'(t) u(t) + \lambda(t) u'(t)$

(c) $\frac{d}{dt} (u(t) \cdot v(t)) = u'(t) \cdot v(t) + u(t) \cdot v'(t)$

(d) $\frac{d}{dt} (u(t) \times v(t)) = u'(t) \times v(t) + u(t) \times v'(t)$

(e) $\frac{d}{dt} (u(\lambda(t))) = \lambda'(t) u'(\lambda(t))$

(f) At point $u(t) \neq 0$ $\frac{d}{dt} |u(t)| = \frac{u(t) \cdot u'(t)}{|u(t)|}$

Ex 7 Show that the speed of a moving particle remains constant over an interval of time if and only if the acceleration is perpendicular to velocity.

$$|v| = \text{constant} \Leftrightarrow |v|^2 = \text{constant} \Leftrightarrow v \cdot v = \text{constant}$$

$$\Leftrightarrow \frac{d}{dt} (v \cdot v) = 0 \Leftrightarrow \frac{dv}{dt} \cdot v + v \cdot \frac{dv}{dt} = 0 \Leftrightarrow 2 \underbrace{\frac{dv}{dt} \cdot v}_{(8)} = 0$$

DEFINITIONS Velocity, Direction, Speed, Acceleration

(Summary)

If $\mathbf{r}$ is the position vector of a particle moving along a smooth curve in space, then

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$$

is the particle's velocity vector, tangent to the curve. At any time t , the direction of $\mathbf{v}$ is the direction of motion. The magnitude of $\mathbf{v}$ is the particle's speed, and the derivative $\mathbf{a} = d\mathbf{v}/dt$, when it exists, is the particle's acceleration vector. In summary,

- ① Velocity is the derivative of position $\mathbf{v} = \frac{d\mathbf{r}}{dt}$
- ② Speed is the magnitude of velocity $\text{speed} = |\mathbf{v}|$
- ③ Acceleration is the derivative of velocity $\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$
- ④ The unit vector $\mathbf{v}/|\mathbf{v}|$ is the direction of motion.

CURVES AND PARAMETRIZATION

→ We will regard a curve in 3-space as the set of points whose positions are given by the position vector function

$$\mathbf{r} = \mathbf{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k} \quad a \leq t \leq b$$

The parameter t , here, need no longer represent time or any other specific quantity

Ex Use $t=y$ to parametrize the part of line of intersection of the two planes $y = 2x - 4$ and $z = 3x + 1$ from $(2, 0, 7)$ to $(3, 2, 10)$

$$y = t$$

$$y = 2x - 4 \Rightarrow x = \frac{y+4}{2} \quad x = \frac{t+4}{2} = \frac{t}{2} + 2$$

$$z = 3x + 1 \Rightarrow z = 3\left(\frac{t}{2} + 2\right) + 1 = \frac{3t}{2} + 7$$

$$\mathbf{r}(t) = \frac{t+4}{2}\vec{i} + t\vec{j} + \left(\frac{3}{2}t + 7\right)\vec{k} \quad \text{from } (2, 0, 7) \text{ to } (3, 2, 10) \\ 0 \leq t \leq 2.$$

Ex 7 The plane $x+y=1$ intersects the paraboloid $z=x^2+y^2$ in a parabola. Parametrize the whole parabola using $t=x$ as parameter.

Could $t=y$ have been used as a parameter?

Parabola $x, y = 1-x, z = x^2+y^2$

$x=t$ $y=1-t$ $z=t^2+(1-t)^2 = t^2+1-2t+t^2 = 2t^2+1-2t$

Parabola $r(t) = t\vec{i} + (1-t)\vec{j} + (1-2t+2t^2)\vec{k} \quad -\infty < t < \infty$

$y=t$ $y=t$ $x=1-t$ $z=t^2+(1-t)^2 = t^2+1-2t+t^2 = 2t^2+1-2t$

Parabola $r(t) = (1-t)\vec{i} + t\vec{j} + (1-2t+2t^2)\vec{k} \quad -\infty < t < \infty$

Ex 7 show that each of the vector functions

① $r_1(t) = \sin t \vec{i} + \cos t \vec{j} \quad (-\pi/2 \leq t \leq \pi/2)$

② $r_2(t) = (t-1)\vec{i} + \sqrt{2t-t^2}\vec{j} \quad (0 \leq t \leq 2) \text{ and}$

③ $r_3(t) = t\sqrt{2-t^2}\vec{i} + (1-t^2)\vec{j} \quad -1 \leq t \leq 1$

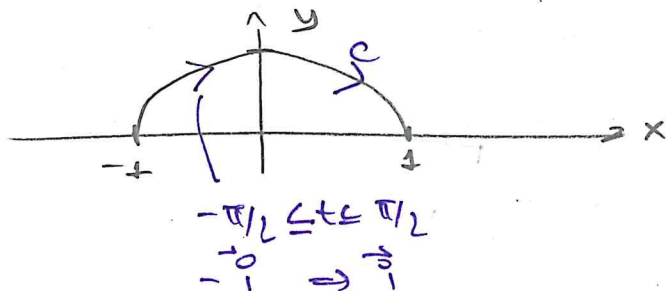
all represent the same curve. Describe the curve

① $r_1(-\pi/2) = -\vec{j} \quad r_1(\pi/2) = \vec{j}$

$\cos t \geq 0 \quad -\pi/2 \leq t \leq \pi/2$ upper half of the xy plane.

$|r_1(t)| = \sqrt{(\sin t)^2 + (\cos t)^2} = 1$

$y = \sqrt{1-x^2}$



② $r_2(0) = -\vec{i}$
 $r_2(2) = \vec{i}$

$|r_2(t)| = \sqrt{(t-1)^2 + (2t-t^2)} = \sqrt{\cancel{t^2} - 2\cancel{t} + 1 + 2\cancel{t} - \cancel{t^2}} = 1$
 $\sqrt{2t-t^2} = \sqrt{1 - (t-1)^2} = \sqrt{1-x^2}$

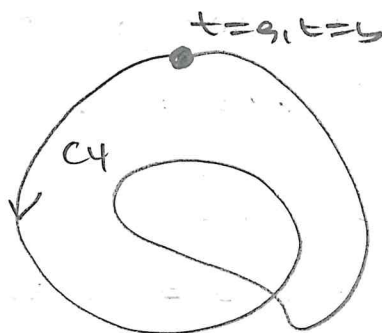
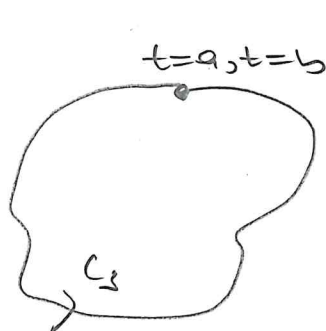
③ $r_3(-1) = -\vec{j}$
 $r_3(1) = \vec{j}$

$|r_3(t)| = \sqrt{t^2(2-t^2) + (1-t^2)^2}$
 $= \sqrt{2t^2 - \cancel{t^4} + 1 - 2t^2 + \cancel{t^4}} = 1$

$y = \sqrt{1-x^2}$

$(1-t^2) = \sqrt{1 - (t\sqrt{2-t^2})^2}$
 $= \sqrt{1 - t^2(2-t^2)} = \sqrt{1 - 2t^2 + t^4}$

→ The curve $r=r(t)$ ($a \leq t \leq b$) is called a closed curve if $r(a)=r(b)$, that is, if the curve begins and ends at the same point

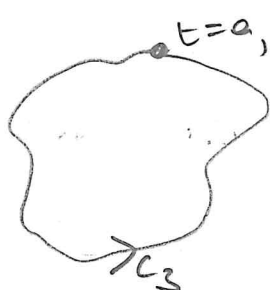
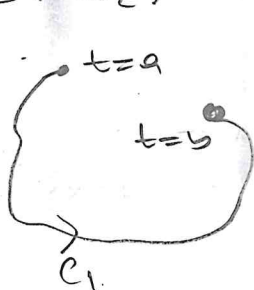


→ The curve C is non-self-intersecting if there exists some parametrization

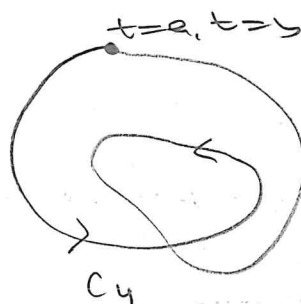
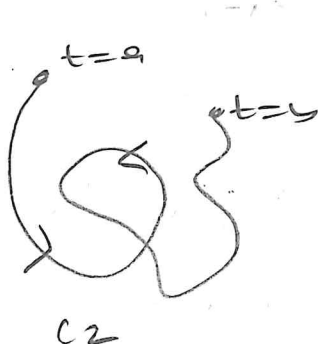
$$r=r(t) \quad (a \leq t \leq b)$$

of C that is one-to-one except the end-points could be the same

$$r(t_1) = r(t_2) \quad a \leq t_1 < t_2 \leq b \Rightarrow t_1 = a \text{ and } t_2 = b$$



non-self intersecting curves



Curves C_2, C_4 intersect themselves

★★

Ex Find parametric equation for the tangent line to the curve $r(t) = t^5 \vec{i} + t^4 \vec{j} + t^3 \vec{k}$ at point $(1,1,1)$

$$\vec{r}(t) = \langle t^5, t^4, t^3 \rangle$$

$$\vec{r}'(t) = \langle 5t^4, 4t^3, 2t^2 \rangle$$

$$\vec{r}'(t) = \langle 5t^4, 4t^3, 2t^2 \rangle$$

Tangent line

$$\langle x, y, z \rangle = \langle 1, 1, 1 \rangle + t \langle 5, 4, 3 \rangle$$

or
(11)

$$\frac{x-1}{5} = \frac{y-1}{4} = \frac{z-1}{3}$$

Ex 7 Parametrize the curve of intersection of the plane

$$x+2y+4z=4 \text{ and the elliptic cylinder } x^2+4y^2=4$$

Sol We begin with the equation $x^2+4y^2=4$ which is independent of z . It can be parametrized in many ways, one convenient way is

$$x=2\cos t \quad y=\sin t \quad 0 \leq t \leq 2\pi$$

Solve the system $x+2y+4z=4$ for z

$$z = \frac{1}{4}(4-x-2y) = 1 - \frac{1}{2}(\cos t + \sin t)$$

Thus the given surfaces intersect the curve

$$\vec{r}(t) = 2\cos t \vec{i} + \sin t \vec{j} + \left(1 - \frac{\cos t + \sin t}{2}\right) \vec{k} \quad 0 \leq t \leq 2\pi$$

Ex 7 Find a parametric representation of the curve of intersection of the two surfaces

$$x^2+y+z=2 \text{ and } xy+z=1$$

Let $x=t$ then we have $x^2+y+z=2 \wedge xy+z=1$

$$z=1-xy. \text{ Then, we get } x^2+y+1-xy=2$$

$$x^2+y-xy=1 \Rightarrow \begin{matrix} x=t \\ t^2+y(1-t)=1 \end{matrix} \quad y = \frac{1-t^2}{1-t} = 1+t$$

$$z = 1-xy = 1-t \cdot (1+t) = 1-t-t^2$$

Thus, a possible parametrization of the curve is

$$\vec{r}(t) = t\vec{i} + (1+t)\vec{j} + (1-t-t^2)\vec{k}$$

INTEGRALS

Definition The definite integrals of a continuous function

$$\vec{r}(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k}$$

$$\int_a^b \vec{r}(t) dt = \left(\int_a^b f(t) dt \right) \vec{i} + \left(\int_a^b g(t) dt \right) \vec{j} + \left(\int_a^b h(t) dt \right) \vec{k}$$

Ex: If $\vec{r}(t) = 2\cos t \vec{i} + \sin t \vec{j} + 2t \vec{k}$ then $\int_0^{\pi/2} \vec{r}(t) dt = ?$

$$\begin{aligned} \int_0^{\pi/2} \vec{r}(t) dt &= \left(\int_0^{\pi/2} 2\cos t dt \right) \vec{i} + \left(\int_0^{\pi/2} \sin t dt \right) \vec{j} + \left(\int_0^{\pi/2} 2t dt \right) \vec{k} \\ &= 2\sin t \Big|_0^{\pi/2} \vec{i} - \cos t \Big|_0^{\pi/2} \vec{j} + t^2 \Big|_0^{\pi/2} \vec{k} \\ &= 2\vec{i} + \vec{j} + \frac{\pi^2}{4} \vec{k} \end{aligned}$$

ARC LENGTH

→ Let C be a bounded, continuous curve specified by

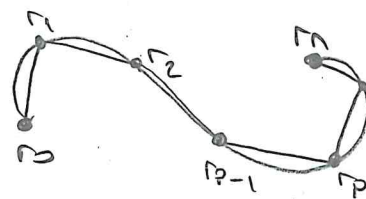
$$\vec{r} = \vec{r}(t) \quad a \leq t \leq b$$

Subdivide the closed interval $[a, b]$ into n subintervals by points

$$a = t_0 < t_1 < t_2 < \dots < t_{n-1} < t_n = b$$

$$S_n = \sum_{i=1}^n |\vec{r}_i - \vec{r}_{i-1}| = \sum_{i=1}^n \left| \frac{\Delta \vec{r}_i}{\Delta t_i} \right| \Delta t_i$$

$$\begin{aligned} s &= \lim_{\substack{n \rightarrow \infty \\ \max \Delta t_i \rightarrow 0}} S_n = \int_a^b \left| \frac{d\vec{r}}{dt} \right| dt \\ &= \int_a^b |\vec{v}(t)| dt \end{aligned}$$



Recall that for $x = f(t)$ $y = g(t)$ $a \leq t \leq b$

$$L = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2} dt = \int_a^b \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt$$

$$\vec{r}(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k}$$

$$L = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} dt = \int_a^b |\vec{r}'(t)| dt$$

Ex 7 Find the length s of the circular helix

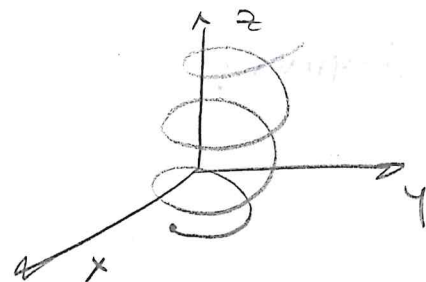
$$\mathbf{r} = a \cos t \vec{i} + a \sin t \vec{j} + b t \vec{k} \quad \text{between points}$$

$$(a, 0, 0) \text{ and } (a, 0, 2\pi b)$$

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = -a \sin t \vec{i} + a \cos t \vec{j} + b \vec{k}$$

$$S = \int_0^{2\pi} |\mathbf{v}(t)| dt = \int_0^{2\pi} \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + b^2} dt$$

$$= \int_0^{2\pi} \sqrt{a^2 + b^2} dt = 2\pi \sqrt{a^2 + b^2}$$



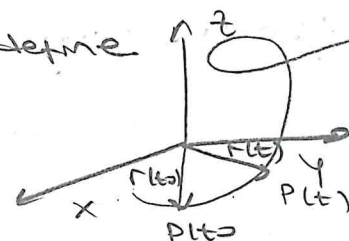
Def Given a smooth curve

$$\mathbf{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k} \quad a \leq t \leq b$$

and a number t_0 ($a \leq t_0 \leq b$) we define

$$s(t) = \int_{t_0}^t |\mathbf{v}(s)| ds$$

is called an arc length parameter for the curve



Ex 7 Parametrize the circular helix

$$\mathbf{r} = a \cos t \vec{i} + a \sin t \vec{j} + b t \vec{k}$$

in terms of the arc length measured from the point

$(a, 0, 0)$ in the direction of increasing t .

$(a, 0, 0)$ corresponds to $t = 0$

$$S = s(t) = \int_0^t \sqrt{a^2 + b^2} ds = \sqrt{a^2 + b^2} t$$

Therefore, $t = s / \sqrt{a^2 + b^2}$ and the arc length parametrization

is

$$\mathbf{r} = a \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \vec{i} + a \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \vec{j} + \frac{bs}{\sqrt{a^2 + b^2}} \vec{k}$$

The Unit Tangent Vector

Let $\mathbf{r}(t)$ be a curve. We assume $\mathbf{v}(t) = \left| \frac{d\mathbf{r}}{dt} \right| \neq 0$ for all t .

$$\hat{\mathbf{T}}(t) = \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{d\mathbf{r}/dt}{|d\mathbf{r}/dt|} \rightarrow \text{the unit tangent vector.}$$

Ex 7 Find the unit tangent vector for the circular helix

$$\mathbf{r} = a \cos t \vec{i} + a \sin t \vec{j} + bt \vec{k}$$

$$\mathbf{v}(t) = -a \sin t \vec{i} + a \cos t \vec{j} + b \vec{k}$$

$$\mathbf{T}(t) = \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{-a \sin t \vec{i} + a \cos t \vec{j} + b \vec{k}}{\sqrt{(a \sin t)^2 + (a \cos t)^2 + b^2}}$$

$$\mathbf{T}(t) = \frac{-a \sin t}{\sqrt{a^2 + b^2}} \vec{i} + \frac{a \cos t}{\sqrt{a^2 + b^2}} \vec{j} + \frac{b}{\sqrt{a^2 + b^2}} \vec{k}$$

in terms of arc-length parameter

$$\mathbf{r}(s) = a \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \vec{i} + a \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \vec{j} + \frac{bs}{\sqrt{a^2 + b^2}} \vec{k}$$

$$\begin{aligned} \vec{T}(s) = \frac{d\mathbf{r}}{ds} &= -\frac{a}{\sqrt{a^2 + b^2}} \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \vec{i} \\ &+ \frac{a}{\sqrt{a^2 + b^2}} \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \vec{j} \\ &+ \frac{b}{\sqrt{a^2 + b^2}} \vec{k} \end{aligned}$$

